

CalibAgent: Task-Aware Active Calibration and Shift Recovery for Quadruped Velocity Commands

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Abstract—Commercial and learned quadruped controllers often expose a planar velocity command while hiding the dynamics that map commands to realized motion. Payload, surface, controller, and estimator changes can therefore preserve balance yet distort navigation. We present CALIBAGENT, an active calibration layer for this opaque interface. A Bayesian command-to-motion model represents cross-axis coupling and asymmetric response, and a task-weighted integrated variance objective selects commands for a declared deployment distribution. Hard authorization, validation stopping, bounded inverse compensation, and residual-triggered recovery remain separate from acquisition. On 183 passive Unitree Go2 trials, a coupled affine map reduces held-out velocity RMSE by 54.5% relative to raw commands. Across registered synthetic families, task weighting reaches a joint accuracy–uncertainty target in 18.67 trials, saving 39.5% over Latin hypercube sampling and 18.8% over D-optimal design. A candidate-noise oracle sensitivity changes the heteroscedastic result by one trial without changing the method ranking. Pinned Isaac Lab experiments show 9.3–31.7% held-out RMSE reduction, positive early-recovery effects under four shifts with no pre-shift alarms, and failure-aware navigation noninferiority across six prospectively frozen maps. The evidence establishes the active mechanism in registered simulation and the model need on passive hardware; the frozen active-hardware study remains to be executed.

I. INTRODUCTION

A quadruped can remain upright while executing the wrong motion. High-level controllers accept a desired planar velocity $\mathbf{u} = [v_x, v_y, \omega_z]^\top$, but the steady body velocity may contain gain error, dead zones, saturation, and cross-axis coupling. These errors are particularly consequential when a planner assumes that the exposed command is a calibrated control variable. Taouil *et al.* showed that learned motion models of a black-box quadruped controller can improve footstep planning [1]; related work has calibrated velocity-controlled kinematic models online for wheeled visual-inertial odometry [2]. The remaining question is how to collect the calibration trials themselves when hardware time is limited and the useful part of command space is task dependent.

Robot calibration has long treated measurement configuration as a design variable [3], [4]. Classical observability and D-optimal criteria prioritize parameter identification [5], [6]. Task-oriented calibration instead scores a design by uncertainty at the end-effector or another quantity of interest [7], [8]. We apply that principle to the external velocity interface of a fixed quadruped controller: trials are chosen for their reduction of predictive uncertainty over commands expected during deployment, not for uniform parameter-space coverage.

This layer differs from locomotion-policy adaptation. RMA conditions a trained policy on latent environment estimates [9], residual models can update model-based legged controllers online [10], and recent embodiment adaptation identifies hardware changes from short interaction histories [11]. CALIBAGENT neither retrains nor instruments the low-level controller. It identifies the mapping from user commands to measured steady motion, translates desired velocities through a constrained inverse, and reactivates calibration when predictive residuals indicate a shift.

This paper makes four contributions. First, it formulates the opaque command-to-motion interface as a compact Bayesian regression with coupled, asymmetric features and per-trial measurement uncertainty. Second, it gives the conditional one-step task-weighted integrated variance reduction (IVR) score and a safe finite-pool sequential design. Third, it composes acquisition with hard authorization, validation stopping, constrained inversion, and latched shift recovery without allowing statistical value to relax an execution limit. Fourth, it evaluates the full argument from physical model need to downstream navigation using passive Go2 data, controlled synthetic families, fault injection, pinned physics simulation, and a retained failed confirmation followed by a disjoint prospective replication. The methodological contribution is this task-measure-to-command design and its closed-loop operational composition; the regression, optimal-design update, and change detector remain individually recognizable components. The current hardware evidence is passive, and the active hardware protocol is frozen but not included in the reported results.

II. RELATED WORK

A. Calibration and Experimental Design

Kinematic and dynamic robot calibration depend on both model structure and the poses or trajectories used for identification [3], [4]. Sun and Hollerbach linked robot observability indices to alphabetic optimal design and developed an efficient active pose-selection algorithm [5], [6]. Carrillo *et al.* then selected configurations by predicted task-space error rather than only parameter covariance [7]. This task-space view parallels goal-oriented Bayesian design, which minimizes uncertainty in a downstream quantity of interest [8]. Greedy information selection also has established connections to near-optimal sensor placement [12]. CALIBAGENT uses a related predictive objective, but its design variable is a safety-filtered velocity command, its integration measure is the planner-facing command distribution, and its posterior is consumed by the same bounded inverse and

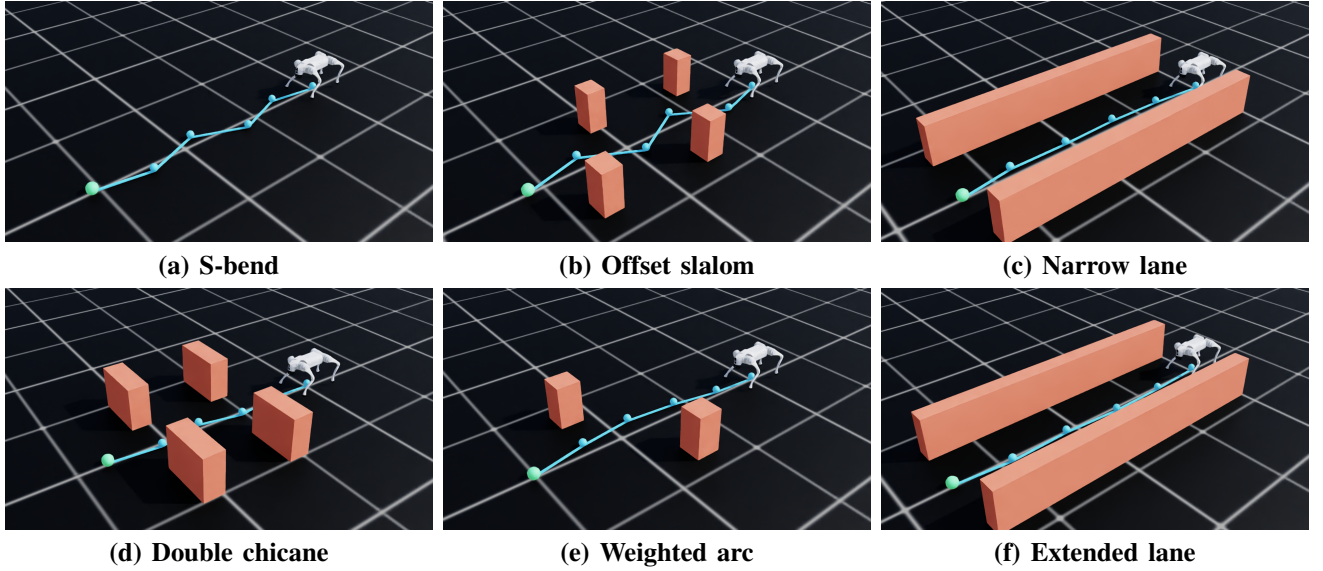


Fig. 1. Elevated Isaac Sim views of all six frozen P7 prospective-replication maps. Cyan segments and spheres reproduce the map waypoints; orange geometry reproduces the registered obstacles. The overlays are non-colliding visualization geometry and do not contribute to the quantitative endpoints.

change detector. In optimal-design terminology, Eq. (4) is a sequential Bayesian I/V-optimal criterion specialized to a multi-output command interface. Its contribution is the task-measure-to-command specialization, the separation of statistical selection from hard authorization, and the end-to-end calibration–recovery evidence.

B. Legged Command Models and Adaptation

Simulation-trained policies have produced agile and terrain-robust quadruped locomotion [13], [14], [15]. Robust locomotion, however, does not make the user-level velocity interface exact. Learned black-box motion models can bridge that interface for planning [1], while steady-state and local dynamics models can adapt a legged robot to unmodeled disturbances [16]. Policy-level adaptation and domain randomization address broader changes [9], [17], [11]. CALIBAGENT occupies a narrower operational layer: it actively calibrates a fixed controller’s command response, quantifies posterior uncertainty, and applies compensation only inside a non-learned envelope.

III. METHOD

Figure 2 separates the probabilistic calibration loop from the rules that constrain its operation. The low-level locomotion controller is fixed and treated as a black box. A trial executes a constant body-frame command after warm-up and ramp-in, extracts a robust steady-motion observation, and records its covariance and validity. Invalid observations never update the posterior.

A. Bayesian Command-to-Motion Model

Let $\mathbf{y} \in \mathbb{R}^3$ denote the measured steady velocity for command $\mathbf{u}$. M1 uses $[1, v_x, v_y, \omega_z]$. M2 has 13 terms: M1, the three pairwise products, and $[v_k - \tau_k]_+$, $[-v_k - \tau_k]_+$ for each axis, with $\tau = [0.15, 0.10, 0.25]$. The non-intercept columns are standardized once on a predeclared safe command pool,

never on observations or held-out outcomes. P1 and P7 use the four-term M1; P3, P5, and P6 use M2. P5–P7 initialize the linear projection to the identity interface and regularize deviations with the frozen prior in Table I; consequently, the 12-trial P7 fit is a four-parameter-per-output affine update, not an underdetermined M2 fit.

For output axis a , the model is

$$\mathbf{y}_a = \boldsymbol{\theta}_a^\top \boldsymbol{\phi}(\mathbf{u}) + \epsilon_a, \quad \epsilon_a \sim \mathcal{N}(0, \sigma_a^2 + r_a), \quad (1)$$

where r_a is the measurement-pipeline covariance diagonal and σ_a^2 is the frozen process variance. Given prior $\mathcal{N}(\mathbf{m}_{a0}, \boldsymbol{\Sigma}_{a0})$, sequential updates use

$$\boldsymbol{\Lambda}_a \leftarrow \boldsymbol{\Lambda}_a + \frac{\boldsymbol{\phi}\boldsymbol{\phi}^\top}{\sigma_a^2 + r_a}, \quad (2)$$

$$\boldsymbol{\eta}_a \leftarrow \boldsymbol{\eta}_a + \frac{\boldsymbol{\phi}\mathbf{y}_a}{\sigma_a^2 + r_a}, \quad \boldsymbol{\Sigma}_a = \boldsymbol{\Lambda}_a^{-1}, \quad \mathbf{m}_a = \boldsymbol{\Sigma}_a \boldsymbol{\eta}_a. \quad (3)$$

The predictive mean is $\mu_a(\mathbf{u}) = \mathbf{m}_a^\top \boldsymbol{\phi}(\mathbf{u})$, and the predictive variance is $\boldsymbol{\phi}^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi} + \sigma_a^2$. Independent output posteriors keep the online update small while coupled input features represent cross-axis response. Aggregate error uses $\|\mathbf{e}\|_W^2 = \mathbf{e}^\top \mathbf{W}^\top \mathbf{W} \mathbf{e}$ with $\mathbf{W} = \text{diag}((1 \text{ m s}^{-1})^{-1}, (1 \text{ m s}^{-1})^{-1}, (1 \text{ rad s}^{-1})^{-1})$; this is numerically the registered equal-axis convention. Axiswise errors are reported alongside the scalar score.

B. Task-Weighted Active Selection

A frozen task distribution is represented by grid commands $\{\mathbf{g}_j\}_{j=1}^G$ and normalized weights w_j . For candidate $\mathbf{c}$, let $\tilde{r}_a(\mathbf{c})$ denote measurement variance available at planning time. A Sherman–Morrison update gives the outcome-independent, conditional one-step reduction in weighted epistemic variance:

$$\mathcal{I}(\mathbf{c}) = \sum_{a=1}^3 \sum_{j=1}^G w_j \frac{(\boldsymbol{\phi}(\mathbf{g}_j)^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\mathbf{c}))^2}{\sigma_a^2 + \tilde{r}_a(\mathbf{c}) + \boldsymbol{\phi}(\mathbf{c})^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\mathbf{c})}. \quad (4)$$

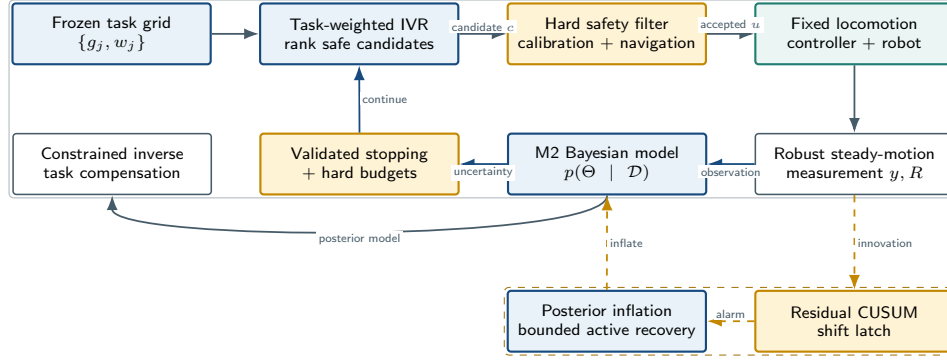


Fig. 2. CALIBAGENT architecture. Task-weighted acquisition ranks calibration commands, but the hard filter alone authorizes execution. Valid observations update the Bayesian model; held-out validation and uncertainty control stopping. A latched residual alarm inflates posterior uncertainty and opens a bounded active-recovery phase. The inverse compensator uses the same bounded command set during navigation.

The registered planner sets $\tilde{r}_a = 0$ because a future trial covariance is not observed before execution; Eq. (3) still uses its realized r_a . When a candidate-noise predictor is available, the implementation accepts its nonnegative $\tilde{r}_a(c)$. The supplemental oracle sensitivity supplies the known synthetic value. The planner maximizes $\mathcal{I}(c) - \lambda_r R(c) - \lambda_d D(c)$, where R is normalized command magnitude and D is normalized linear execution distance, over a bounded finite pool; the hard filter authorizes a candidate only after statistical ranking. P3 freezes both coefficients at zero to isolate IVR. Greedy batches use covariance-only fantasy updates, so no unobserved target is imputed. P3 uses 256 candidates, a six-command seed design, a 256-point task grid, a disjoint 400-point evaluation grid, and a 0.025 normalized duplicate radius.

C. Safety, Stopping, Compensation, and Shift Recovery

The hard filter checks finite state, localization, battery, roll, pitch, base height, command bounds, linear norm, coupled linear-angular load, command slew, and projected workspace occupancy. It evaluates ranked candidates after planning and fails closed when none remains. A 50 Hz monitor can zero the command independently of the planner. This separation prevents uncertainty reduction from trading against a learned safety penalty.

Stopping requires minimum trial coverage and held-out validation. A run stops at the registered target only after three consecutive confirmations of both validation RMSE and integrated uncertainty; a low-gain condition also requires valid held-out accuracy. Trial, time, distance, and battery budgets take priority. For deployment, the inverse compensator searches the same bounded candidate set and minimizes

$$\|W(\mu(u) - v^*)\|_2^2 + \lambda \|W(u - v^*)\|_2^2 + \rho \sum_a s_a^2(u), \quad (5)$$

then passes the selected command through the hard filter.

Shift detection uses normalized innovation energy $z_t = e_t^\top S_t^{-1} e_t / 3$. A one-sided CUSUM accumulates $q_t = \max(0, q_{t-1} + z_t - 1 - \kappa)$ and alarms only after a frozen threshold, minimum dwell, and repeated positive evidence

in a bounded window. The alarm latches, inflates epistemic covariance without changing the posterior mean, and starts a capped recovery budget. Normal stopping resumes after recovery.

IV. EXPERIMENTAL PROTOCOL

The evaluation follows a staged evidence ladder; Table I states the inferential unit, partitions, timing, and gates. P1 contains 183 valid passive Unitree Go2 trials in three 61-trial sessions. Livox MID360 FAST-LIO/global localization supplies the reference pose. Each complete session is held out once; 30 selected trials train sampled designs and the other 122 form the dense reference. P1 tests physical model structure, not active execution.

P3 is a controlled sample-efficiency benchmark. Twenty independent seeds span affine, dead-zone, and heteroscedastic repeated conditions; inference first averages families within seed. Candidate, task, and evaluation Sobol sets use independent seeds 31013, 11003, and 23011. The evaluation grid is an offline ground-truth benchmark and never enters acquisition. A second 1,024-command audit grid (seed 43019) evaluates each posterior once at its primary crossing. Controls are random, LHS, Sobol, Bayesian D-optimal, no-task IVR, and a 256-command oracle. P4 then replays the 60 frozen active traces through the operational stopping rule and injects proposal/state faults.

P5–P7 use Isaac Lab v2.3.2 at commit 37ddf62, Isaac Sim 5.1.0-rc.19, and PhysX [18]. P5 tests four response contexts with 20 paired seeds. P6 tests four held-out in-place shifts with 72 paired seeds and frozen, passive-update, and full-recovery controls. Its first four monitor trials are a no-shift null; the registered early window is recovery trials 4–9. P7 has six disjoint maps, seven methods, and 72 new paired seeds per map (3,024 episodes). All arms share the same controller checkpoint, 50 Hz monitor and interlock, 10 Hz waypoint follower, map geometry, seed, and planner-configuration hash; only calibration differs. Rate endpoints use exact binomial intervals and continuous paired endpoints use 4,000-sample seed bootstrap intervals.

TABLE I

FROZEN EXPERIMENT AND ENDPOINT SPECIFICATION. A “TRIAL” IS ONE CONSTANT BODY-FRAME COMMAND FOLLOWED BY ONE STEADY-MOTION OBSERVATION. ALL BUDGETS INCLUDE THEIR SEED COMMANDS.

Stage	Independent unit	Model and budget	Partition / timing	Primary endpoint and frozen rule
P1	Acquisition session ($n = 3$)	M0/M1; sampled 30, dense 122	Whole-session ramp/settle/measure/ramp-out 0.6/0.8/2.0/0.6 s	holdout; Held-out three-axis and per-axis RMSE; sampled M1 versus raw and M0
P3	Seed ($n = 20$); three repeated families	M2 (13 terms); six-command seed; horizon 160	Candidate/task/evaluation/audit sets mutually seeded; outcomes never enter design	First trial with RMSE ≤ 0.04 and integrated epistemic variance ≤ 0.0015 ; 20-seed paired inference
P4	Frozen P3 trace ($n = 60$); injected case	Three-confirmation stop; 300 proposal hazards; 160 sampled-state faults	Bounds $ v_x \leq 0.75$, $ v_y \leq 0.45$, $ \omega_z \leq 1.2$; monitor period 20 ms	No premature stop; overshoot; rejection/reason coverage. Timing is software decision latency, not actuation latency
P5	Paired simulator seed ($n = 20/\text{context}$)	M2, identity-projection prior scale 0.01; 12 trials	8 fixed validation commands; 1.0/0.4/0.8/1.0/1.0 s trial phases	Raw versus calibrated held-out RMSE; paired absolute effect
P6	Paired simulator seed ($n = 72/\text{shift}$)	M2, identity-projection prior scale 0.01; 12 pre-calibration + 12 recovery	4 no-shift + 5 shifted monitor trials; four-command rolling validation; 0.5/0.3/0.6/0.7/0.5 s phases	False alarm; detection/recovery; passive-minus-full mean RMSE over trials 4–9; terminal upper interval < 0.14
P7	Paired simulator seed ($n = 72/\text{map}$)	M1 (4 terms), identity prior scale 0.10; full/controls 12, dense 30	8 fixed calibration-validation commands; 60 s navigation timeout; 0.25 m way-point/goal radius	Success and collision; 60-s capped completion time (every failure = 60 s); success and time noninferiority both required

The P7 command envelope is $v_x \in [-0.40, 0.40]$ m/s, $v_y \in [-0.30, 0.30]$ m/s, and $\omega_z \in [-0.70, 0.70]$ rad/s, with linear norm at most 0.45 m/s. Cruise, lateral, and yaw limits are 0.18 m/s, 0.14 m/s, and 0.25 rad/s. Success means entering the 0.25 m goal ball without collision or serious safety event before 60 s. Arrival time is defined only on success; the primary failure-aware time estimand assigns 60 s to every failure. Navigation time excludes calibration, which is reported separately: the deterministic executed-command duration is 31.2 s for a 12-trial method and 78.0 s for dense calibration.

V. RESULTS

A. Model Need and Calibration Efficiency

Figure 5a shows a consistent held-out benefit from coupled calibration on the passive Go2 data. Pooled RMSE falls from 0.06605 for raw commands to 0.04563 for diagonal M0 and 0.03009 for coupled M1. M1 therefore reduces error by 54.45% versus raw commands and 34.07% versus M0. The three held-out-session M1 values are 0.02888, 0.03158, and 0.02974. The 30-trial LHS M1 result is 3.33% above the 122-trial dense reference (0.02912). These data support a coupled correction for this robot and acquisition day. Axiswise raw to M1 RMSE is 0.04879 to 0.02587 for v_x , 0.08459 to 0.02569 for v_y , and 0.05960 to 0.03723 for ω_z . The M1-over-M0 gain is concentrated in lateral motion (60.2%); v_x changes by -0.4% and ω_z by 1.0% . Table II shows why: yaw-to-lateral coupling remains 0.119–0.129 in every complete-session holdout. The result measures passive hardware model need, not active hardware efficiency.

P3 isolates the acquisition rule (Fig. 5b). The full method reaches the joint target in 18.67 trials, compared with 30.87 for LHS, 30.67 for random, 25.72 for Sobol, 23.00 for D-optimal, and 25.67 without task weighting. Reductions are 39.52%, 39.13%, 27.41%, 18.84%, and 27.27%, respectively; all one-sided paired Wilcoxon tests give $p = 9.54 \times 10^{-7}$. The primary LHS comparison saves 12.20 trials with a 95% interval of [10.73, 13.73]. Horizon-end active RMSE

TABLE II

P1 M1 GAIN/COUPLING MATRIX. ENTRIES ARE THE MEAN AND RANGE OVER THE THREE LEAVE-ONE-SESSION-OUT FITS; ROWS ARE REALIZED AXES AND COLUMNS ARE COMMAND AXES.

	v_x	v_y	ω_z
v_x	.884 [.878,.890]	.015 [-.005,.043]	-.004 [-.006,-.002]
v_y	.021 [.017,.025]	.653 [.629,.667]	.125 [.119,.129]
ω_z	-.001 [-.007,.003]	-.051 [-.076,-.019]	.914 [.906,.926]

is 0.008462, 6.90% above the dense oracle’s 0.007916, and 95% predictive coverage is 94.14%.

The uncertainty gate is binding in all 60 active conditions, as expected for an uncertainty-directed design; mean accuracy-only crossing occurs at 13.55, 6.40, and 8.80 trials for affine, dead-zone, and heteroscedastic families. Figure 5c therefore reports every family separately. Task IVR requires 18, 18, and 20 trials, versus 19, 19, and 31 for D-optimal and 23, 23, and 31 without task weights. On the disjoint post-hoc audit grid, active posterior RMSE at the primary crossing is 0.01796, 0.02069, and 0.02315; its axiswise 95% coverage ranges from 0.917 to 0.955. Across the nine combinations of RMSE thresholds $\{.03, .04, .05\}$ and uncertainty thresholds $\{.001, .0015, .002\}$, task IVR saves at least 3.33 trials over D-optimal, 6.00 over no-task IVR, and 10.03 over LHS, with a positive paired difference for every seed. Thus the result is not an artifact of a single threshold.

The candidate-noise sensitivity addresses the conditional variance in Eq. (4). In the heteroscedastic family, the registered fixed-noise planner reaches the target in 20 trials; an oracle supplied the exact candidate-dependent excess variance and reaches it in 21. Their crossing RMSEs are 0.02332 and 0.02320. Both retain an 11-trial advantage over D-optimal and no-task IVR. Candidate-noise modeling changes one command in the crossing budget but does not explain the method ranking.

In closed-loop P5 simulation (Fig. 3e), raw-to-calibrated RMSE is 0.15199 to 0.10635 for affine distortion, 0.15054 to 0.10282 for dead-zone distortion, 0.10741 to 0.09742

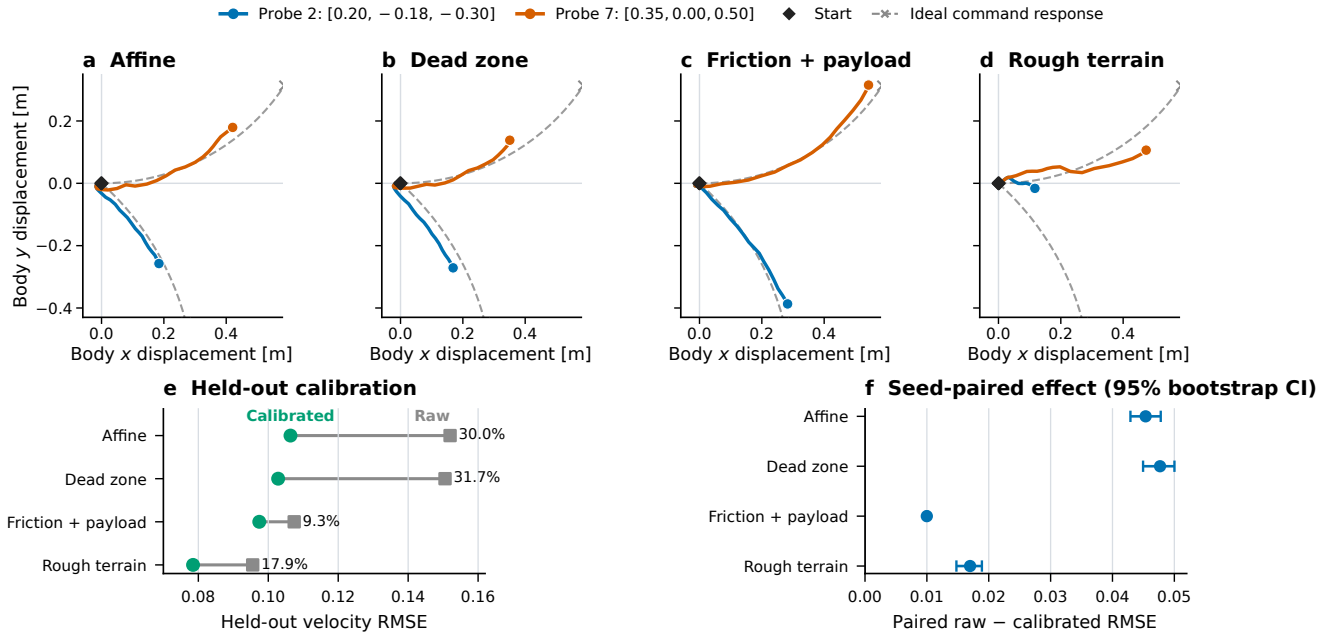


Fig. 3. P5 registered simulation responses and closed-loop calibration. (a)–(d) Body-displacement trajectories for two validation probes at display seed 5301; colored curves are measured, dashed gray curves and crosses are the ideal response to the declared command/timing, and diamonds mark the common start. (e) Raw and calibrated held-out velocity RMSE across 20 paired seeds; labels give relative reduction. (f) Seed-paired absolute improvement with 95% 4,000-sample bootstrap intervals. The trajectories are qualitative examples; all inference uses the registered paired summaries.

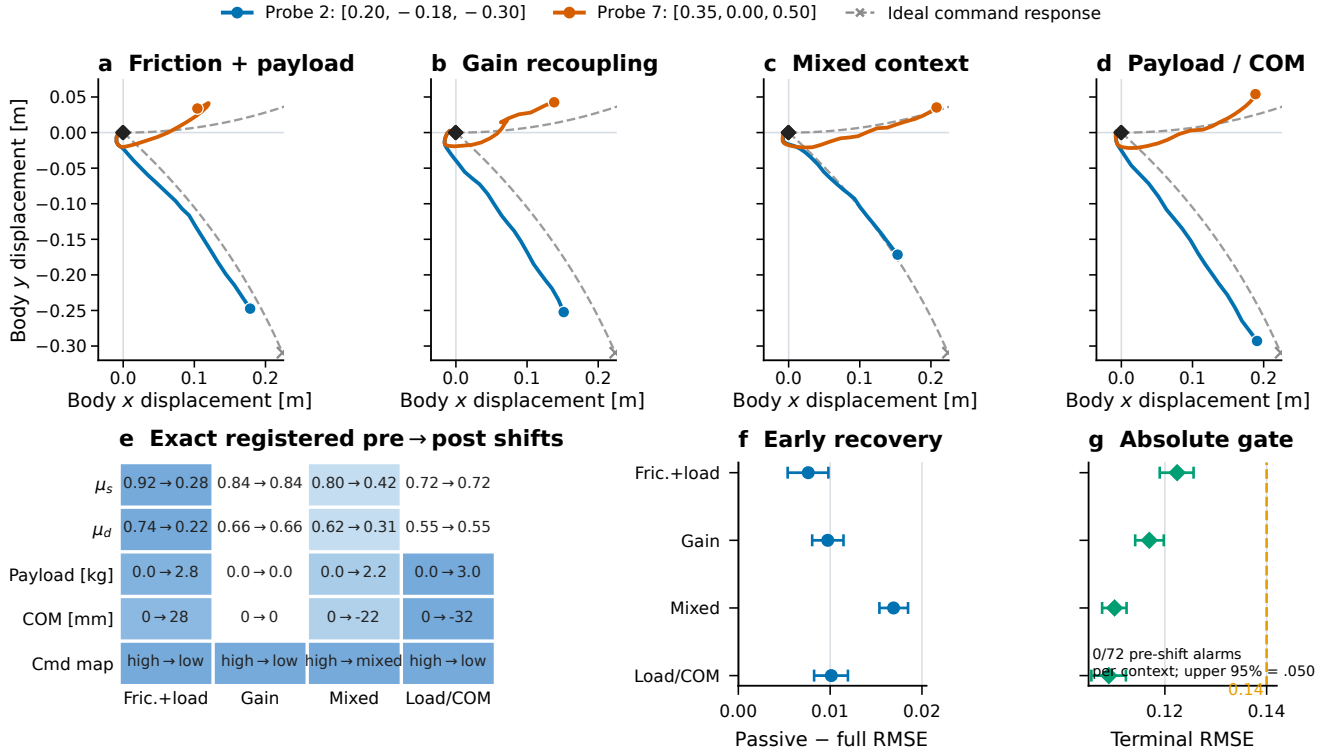


Fig. 4. P6 registered held-out shifts and active recovery. (a)–(d) Measured body-displacement trajectories and dashed ideal command responses for the same validation probes at display seed 10101. (e) Exact pre→post static and dynamic friction, payload, fore-aft COM, and command-map settings; color intensity encodes within-row shift magnitude. (f) Full-method improvement over passive updating in the registered early window. (g) Full-method terminal RMSE and the frozen 0.14 gate. Error bars are paired 95% 4,000-sample bootstrap intervals over 72 seeds per shift; the detector produced 0/72 pre-shift alarms in each context (exact upper bound 0.050).

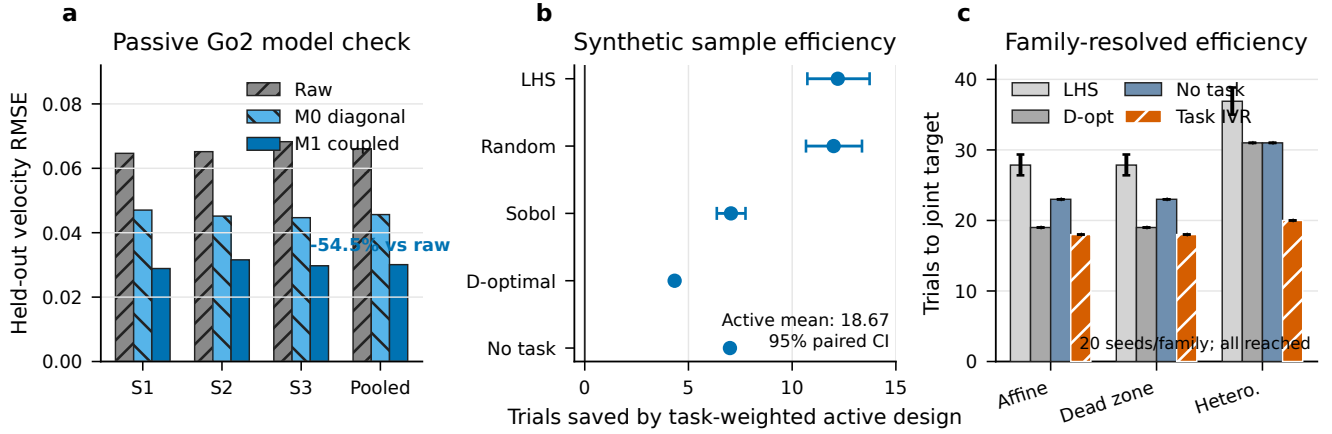


Fig. 5. Calibration evidence. (a) Passive real-Go2 held-out RMSE for raw, diagonal M0, and coupled M1 commands. (b) Paired trials saved by task IVR; bars are 95% seed-bootstrap intervals. (c) Family-resolved time to the same joint target; every method reaches it in all 20 seeds per family.

for low friction plus payload, and 0.09554 to 0.07849 on rough terrain. Relative reductions span 9.30–31.70%. The corresponding paired absolute-improvement intervals are [0.04290, 0.04780], [0.04495, 0.05002], [0.00981, 0.01017], and [0.01477, 0.01889]. All 20 paired seeds improve in each scenario.

B. Stopping, Safety, and Shift Recovery

P4 records no premature stop across 60 trajectories. Median and p95 overshoot after the oracle target are both two trials. The hard filter rejects 300/300 registered hazards and 0/20 safe controls; its sampled-state monitor rejects all 160 hazardous calls at the call in which the invalid state is presented. The earlier 0 ms log value is therefore a software decision timestamp, not a sensor-to-actuator latency measurement. In P5–P7, where the monitor runs against physics state at 50 Hz, the maximum recorded abort interval is 20 ms. Neither result is physical safety certification.

Across the four P6 no-shift windows, the full detector produces 0/72 alarms per context; each exact 95% upper bound is 0.04994. Across the subsequent shifts, the full method improves early RMSE over passive updating by 0.00761 for friction plus payload, 0.00974 for gain recoupling, 0.01691 for mixed context, and 0.01012 for payload/COM (Fig. 4f). Their 95% intervals are [0.00537, 0.00981], [0.00804, 0.01145], [0.01536, 0.01848], and [0.00825, 0.01194]. Detection succeeds in 72/72 seeds for three shifts and 71/72 for payload/COM; full recovery succeeds in 72/72 for every shift. Worst p95 detection and recovery delays are four and six trials, corresponding to at most 10.4 s of monitor-command time and 31.2 s for six adaptation-plus-validation cycles under the frozen trial schedule. Full terminal RMSE ranges from 0.10898 to 0.12241, with every upper interval endpoint below 0.126 (Fig. 4g). Passive updating can be competitive at the terminal window, so the supported contrast is the registered early-recovery effect.

C. Navigation Consequence and Replication

The first P7 strong confirmation did not pass. Full-method success was 68/72 on one map and 50/72 on another, and some completion-time ratio interval upper bounds reached 1.347. Trace inspection exposed a 10 Hz guard blind interval in which base height could cross the hard limit between planner ticks. A 50 Hz predictive height interlock was developed, frozen, and applied to every arm in the prospective replication. The six maps, new seeds, common policy, and planner configuration hash 9affe07d9231 were then frozen. No result from the failed confirmation or intervening pilots is pooled with the replication.

In the disjoint replication, Full succeeds in 72/72 episodes on five maps and 70/72 on weighted arc (Fig. 6a), while raw commands succeed in 0/72 or 1/72. Every Full collision count is zero. The primary time estimand retains all episodes by assigning the 60 s cap to every failure; arrival time among successes is only a diagnostic. Paired raw-minus-Full capped-time gains have lower interval endpoints from 23.453 to 32.675 s (Fig. 6b). The worst Full/dense mean-ratio interval upper bound is 1.0736, below the registered 1.25 margin, and the largest upper bound against any 12-trial matched-budget control is 1.0901 (Fig. 6c). The paired success-difference lower bound also exceeds the registered -0.05 margin in every comparison. The two Full failures are weighted-arc timeouts, not collisions or serious safety events; their final goal distances are 0.565 and 0.567 m.

Navigation time excludes calibration. Full requires 31.2 s of deterministic command execution versus 78.0 s for dense calibration; matched-budget controls have the same 31.2 s budget. Identical seeds, policy, planner, interlock, maps, and navigation logic isolate calibration inside this simulator pipeline. The active-hardware protocol tests the corresponding physical navigation claim.

VI. DISCUSSION AND LIMITATIONS

The experiments separate three questions that are often conflated. P1 asks whether the exposed command interface

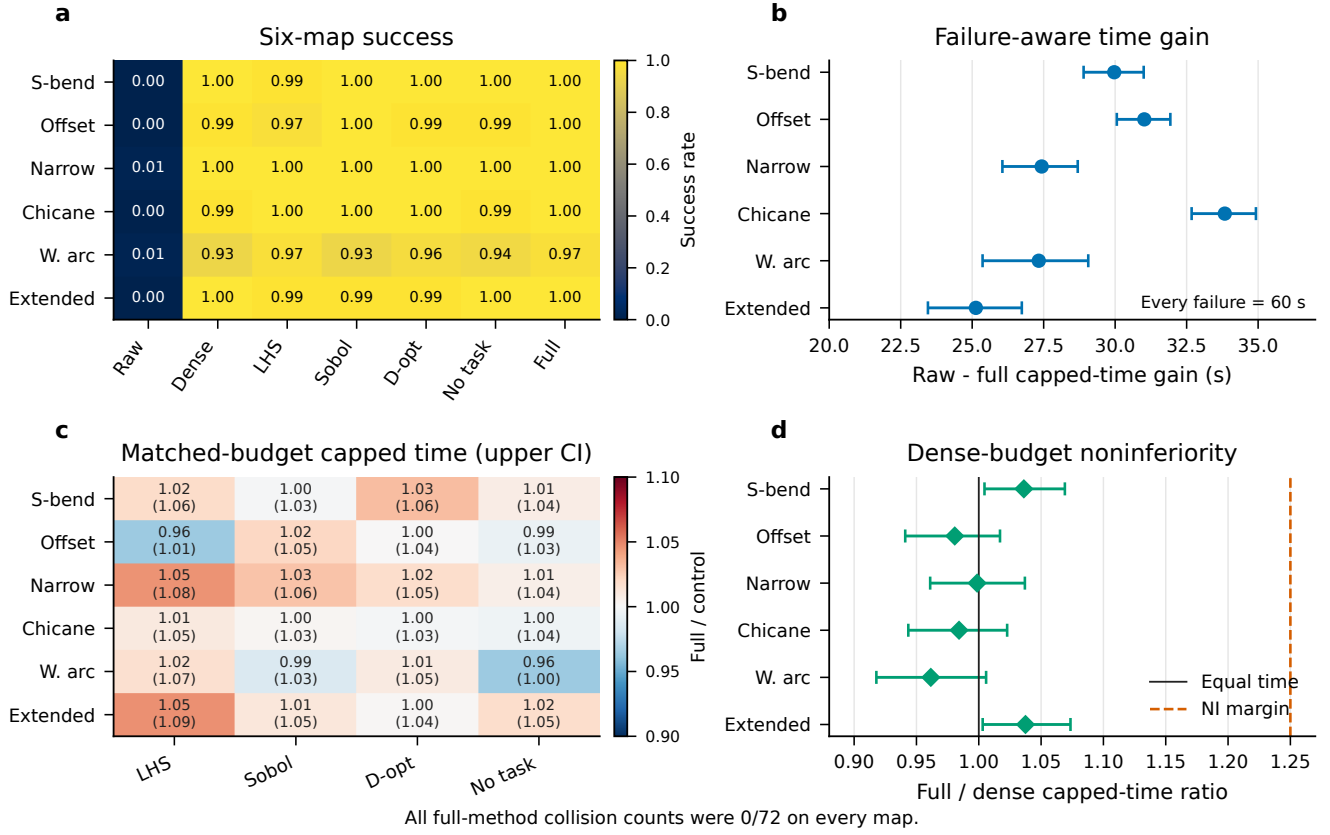


Fig. 6. P7 prospective simulator replication (72 seeds/cell). (a) Success rate. (b) Paired raw-minus-full gain in the 60-s capped-time estimand, which retains every failure as 60 s. (c) Full/matched-budget mean capped-time ratios; parentheses give paired 95% interval upper bounds. (d) Full/dense ratios and the registered 1.25 noninferiority margin. Every full-method collision count is 0/72.

needs coupled calibration. P3 asks whether task weighting changes which trials are valuable. P7 asks whether a calibration that looks accurate on held-out commands also preserves navigation outcomes. The no-task and D-optimal controls isolate task weighting in P3; candidate-noise and threshold sweeps preserve that ordering. In P7, the dense and matched-budget controls establish downstream noninferiority, while the raw arm exposes the consequence of leaving the command interface uncalibrated.

The P3 joint stopping time is uncertainty-limited in all 60 active conditions. This identifies the measured efficiency as faster acquisition of posterior confidence, rather than earlier satisfaction of the accuracy gate alone. The independently seeded 400-point evaluation set supplies the online accuracy gate, and the additional 1,024-point audit verifies RMSE and axiswise coverage at the recorded crossing. P6 similarly separates monitoring from recovery: four pre-shift monitor commands measure false alarms, shifted monitor commands measure detection, and trials 4–9 measure the registered early-recovery effect. The null exposure supports the reported per-context false-alarm bound; long-duration nuisance alarms remain a hardware endpoint.

P1 is passive, same-day data from one Go2 in one environment. P5–P7 use a pinned simulator and fixed policy. The compact steady-response basis targets the steady command

interface; arbitrary history dependence, gait switching, rapid transients, and shifts outside the registered families require a richer model. The frozen P8 study executes active calibration on hardware, applies gain/coupling, payload/COM, friction, and mixed shifts, and tests navigation on weighted-arc and offset-slalom courses. Those experiments complete the active hardware and sim-to-real evidence layer of the same stated mechanism.

VII. CONCLUSION

CALIBAGENT calibrates the high-level velocity interface of a fixed quadruped controller by directing trials toward task-relevant predictive uncertainty. The approach combines a coupled Bayesian model, conditional IVR acquisition, hard safety filtering, validated stopping, constrained inverse compensation, and latched shift recovery. Passive Go2 measurements establish the need for a coupled map; registered synthetic and Isaac Lab experiments establish sample efficiency, early shift recovery, and downstream navigation under controlled conditions. The remaining test is active real-robot execution of the same frozen protocol.

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